An overview study on the use of Semantics in Immersive Environment

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ABSTRACT

This paper provides a comprehensive review of the use of semantic technologies in immersive environments, particularly focusing on Virtual Reality (VR), Augmented Reality (AR), and other forms of Extended Reality (XR). It discusses the role of ontologies and knowledge representation in managing virtual environments (VEs) and highlights the benefits of using knowledge graphs and rule-based reasoning to improve the efficiency and understanding of user interactions, specifically in the ability to generate multimodal commands. By analyzing various approaches and motivations for representing VEs semantically, this review identifies gaps in current research and offers insights into how semantic technologies can be leveraged to create more intuitive and effective immersive experiences. By synthesizing findings from recent studies, the paper aims to guide future research in the field of semantics in immersive environments.

Index Terms: Ontology, Semantic, Knowledge Representation, Immersive Environment, Virtual Reality, Augmented Reality, Mixed Reality, Extended Reality, Virtual Environment, Systematic Review

1 INTRODUCTION

The emergence of Virtual Reality (VR) and Augmented Reality (AR) has opened up new immersive perspectives but has also presented significant challenges in terms of user interface. Unlike traditional computing environments, VR/AR spaces do not offer a keyboard or mouse, leading to the need to intelligently integrate both gesture control mediated by specific interaction devices and voice control. This management of multimodality still raises difficult research problems at the frontier between human-machine interaction and artificial intelligence. Despite significant progress in these two areas, theoretical and technological barriers related to Bolt's paradigm, 'Put this object here' [4], an archetype of all multimodal control, remains one of the main obstacles to the use of virtual and augmented reality in everyday and professional settings. Although progress in deep learning and language models now makes it possible to recognize gestures or voice commands more efficiently, the integration of several modalities to construct a command assumes using in-depth and structured knowledge of both the interaction context and the content of the virtual scene. Neither deep learning nor language models are likely to address this problem globally, which requires a formal, logical, and explicit semantic representation of the concepts handled. We aim to explore the challenges of using semantic data in virtual environments in different application fields and for different motivations, including multimodal interactions, to leverage knowledge representation to address these complexities effectively.

Virtual Environments (VEs) are computer-generated, immersive simulated environments that aim to create a digital twin of real life (e.g., [28, 26, 23]) or entire imagined spaces that can be explored and interacted with by users. These environments utilize diverse technologies to provide sensory feedback, including visual, auditory, and occasionally tactile stimuli. They are employed in multiple different fields such as entertainment, gaming, education [51], healthcare [2], architecture, and urban planning [28], as well as more expert fields like meteorology and biochemistry [47, 41]. Therefore, VEs serve as powerful tools for creativity, exploration, and problem-solving, offering users the freedom to navigate and interact in their immersive environment.

To fully harness the potential of these immersive environments, it is essential to consider how users interact with them. This is where multimodal interaction comes into play. Multimodal interaction refers to the capacity to engage with computers (or other devices) by combining different forms of communication, including speech, gesture, and gaze, in a coordinated manner. In VEs, as stated by Latoschik [32], "multimodal interfaces are promising alternatives to desktop input devices and WIMP (windows, icons, menus, pointer) metaphors", as they enable users to engage with technology more directly. Although first studied by Bolt in his "Put-That-There" paradigm [4] over 44 years ago, realizing multimodal interaction in interactive time remains an unsolved challenge, due to the time required to process the commands because of hardware capabilities and the lack of relation between user interaction and scene content (missing context).

To address these challenges, advances in semantic representation and knowledge representation offer promising solutions. Semantic representation and knowledge representation focus on representing and structuring information, establishing relationships between them to enable machines to understand them. Furthermore, a good data representation enables machines to reason very quickly by using rules to deduce new knowledge. Over the last few decades, numerous technologies have emerged and evolved, considerably improving the reasoning capabilities of knowledge representation systems, and enabling interactive times to be achieved on queries that previously took several seconds. For example, the Semantic Web is an extension of the current web that aims to make data more machine-readable by using standards such as the Resource Description Framework (RDF) and the Web Ontology Language (OWL) to define and structure knowledge and rules, and the SPARQL Protocol And RDF Query Language (SPARQL) can be used to query these graphs to retrieve stored data, as well as inferred data. To facilitate the description of certain generic concepts, ontologies have been created and made available across the web (e.g., GeoSPARQL to represent spaces and deduce information such as intersections, or Time Event Ontology to represent complex temporal relations).

Moreover, a requirement is the ability to logically reason about the concepts describing the interaction context and those relating to the contents handled to consider the user's intentions. When it comes to modelling a virtual simulation, the problem is to represent objects in space during the time, but also to link the interaction with the scene content (What is the user pointing at? looking at? seeing? talking about?), which needs to be interpreted to generate commands (movement, rotation, colour change, etc.). Many different ontological approaches have been proposed to represent the content of a virtual simulation to achieve one or more motivations. Still, so far none of them has been designed to be sufficiently descriptive and general to be recommended and reused in the domain, requiring the wheel to be recreated each time.

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This paper investigates how to effectively represent virtual environment (VE) data for user experience management and multimodality. It explores the advantages of representing the content of VEs in graph knowledge, highlighting how this representation can enhance user interaction and decision. Furthermore, the paper examines the benefits of connecting user interactions with the semantic content of a VE. Additionally, it delves into the contributions of using knowledge graphs and rule-based reasoning in managing interactions within VEs, showing how these technologies can improve the efficiency and understanding of user interactions.

We conducted our literature review according to the guidelines set by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology [38]. We focused on articles from peer-reviewed scientific journals and conferences published over the last twenty years to provide an overview of recent developments in the field of semantics in VEs. We examined the work led in semantics to represent VEs (content and interaction context) and the different motivations and approaches to create such structures. By analyzing the current state of the literature and identifying research gaps, this review will provide insights to efficiently structure data from VEs for user experience (UX) management and interaction multimodality in real-time.

The paper is organized as follows. Section 2 provides the context for the literature review. Section 3 details the approach to address the research questions. The aim is to study the different motivations for representing virtual environments, as well as the different ways of structuring this data to represent VEs and all the reasoning around it. Section 4 presents the overall results of the analysis. Finally, we derive some guidelines and discuss open problems in Sections 5 and 6.

2 Context

By exploring the current context of semantics usage in VEs, we aim to establish the significance and relevance of the research questions within the existing body of knowledge. This section presents broadly the context of the use of semantic data in VEs, and the need to have an uniform ontology and taxonomy to represent VEs efficiently in real-time, including interaction context, event and scene content, especially to perform multimodal interaction by capturing user intent.

Over the last two decades, semantic technologies have evolved considerably, increasing modelling and reasoning capabilities, calculation speed, and storage capacity. This progress makes it possible to find multiple interests in using such technologies and data structures, particularly in the field of Virtual and Augmented Reality. There are multiple motivations for this, such as collaboration, data analysis and annotation support, user experience flexibility and adaptability, multimodal command generation, decision support systems (DSS)... Despite these varied motivations, we often end up representing knowledge (through the creation of more or less complete ontologies) that has already been created by others in the past. This issue arises from the lack of a sufficiently adaptable ontology for virtual environments that can be reused easily. Unfortunately, we frequently have to recreate ontologies from scratch, especially when working with technologies that rely on the reusability and heterogeneity of data for their effectiveness. Having an ontology capable of completely representing an experience in a virtual environment, with the flexibility to incorporate domain-specific concepts, would be a significant advancement. For example, it would address the multimodality issue by enabling the generation of commands and enhancing post-experience analysis by relating highly heterogeneous data.

Multimodality is a complex subject, particularly in the field of virtual and augmented reality. The multimodality deals with user inputs coming from several sensorimotor channels to provide user commands, and deals with outputs providing visual, audio or haptic rendering or feedback :

- **Multimodal output** refers to the fact that the system is designed to provide users with a result using a different format, different stimuli, or involving multisensorial feedback. For example, Ferey et al. [15] made a VR interaction for protein docking with multisensory output, visual rendering to show different representations of proteins, haptic rendering to provide tactile feedback when the proteins collide, and audio rendering for additional cues during the docking process.

- **Multimodal input** (also known as multimodal interaction) is the act of analyzing and interpreting multiple communication channels in real-time [36]. The challenge of multimodal interaction is merging events into interpretable commands performed by the application (multimodal fusion). Oviatt et al. [37], identify two kinds of architectures to realize multimodal interaction can be realized: "ones that integrate signals at the feature level (i.e., "early fusion") and others that integrate information at a semantic level (i.e., "late fusion")".

The use of multimodality features in AR/VR environments facilitates the creation of direct and ecologic interactions, allowing users to communicate and interact with virtual worlds like how they would interact with the real world, without thinking that he is using a technology, thereby enhancing the sense of immersion [1].

Knowledge representation and reasoning (KR&R) techniques have been used in VR/AR for the past couple of decades to incorporate intelligence, building a machine-understandable view of VEs [41] and to understand user intent [5]. Thanks to these interests, Gutierrez et al. [21] and Latoschik [32] proposed the use of KR&R in developing multimodal interactions 20 years ago! However, it was not yet possible to exploit the full potential of these technologies. Now, with the evolution of these techniques, it should be possible to overcome the technical limitations that made real-time multimodal interaction impossible, and that prevented interaction research from progressing 20 years ago.

In this paper, we aim to demonstrate the interest and capabilities to create and integrate an ontology able to represent completely a VE into a graph (with specifics that can be grafted) over time, to serve our objectives, particularly in terms of interaction reasoning and multimodality. However, the motivations for representing VEs, as well as the effective techniques for modelling them, need to be studied in greater detail. By synthesizing the existing literature on this issue, we also aim to identify effective strategies for representing VEs, as well as highlighting the interest in having a semantic approach to manage user experience in VR/AR environments.

3 Methodology

Our literature review was conducted following the PRISMA guidelines [38]. The review protocol was developed to address the research questions identified in the introduction.

3.1 Search Strategy

A systematic literature search was performed using the following electronic databases: ACM Digital Library, IEEE Xplore Digital Library and HAL Open Archive. The search was conducted to find research articles published in English or French between 2004 and 2024. The search strategy was based on a combination of keywords and boolean operators.

3.2 Inclusion and Exclusion Criteria

Articles were included if they met the following criteria:

1. Peer-reviewed articles were published in English or French.

2. Articles on semantic data modelling to represent VEs (content representation).

The search terms used in this review are as follows: **(”Ontology” OR ”Semantic” OR ”Knowledge Representation”) AND (”Immersive Environment” OR ”Virtual Reality” OR ”Augmented Reality” OR ”Mixed Reality” OR ”Extended Reality” OR ”Virtual Environment”)**

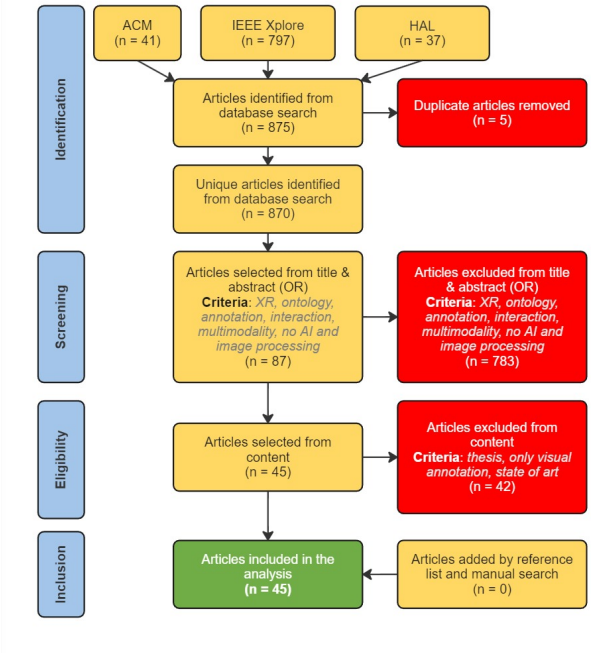


Figure 1: Flow diagram for selection and screening process following PRISMA methodology.

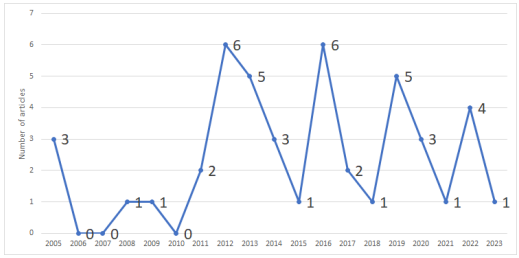


Figure 2: Number of articles published each year on the use of semantic representation for virtual environments.

3. Articles on the use of multimodality and reasoning interaction (interaction context).

Articles were excluded if they did not meet the inclusion criteria or if they were duplicates, posters, or literature reviews.

With the above criteria, we downloaded metadata of 875 articles from three digital libraries and transformed it into CSV format files for further analysis. 5 duplicate articles were removed. From the CSV files, a single file was built in xlsx format with the most important data such as references, year, platform, link to the paper, and others.

3.3 Screening Process

The titles, abstracts, and images of the identified papers were examined to exclude papers that had no information about virtual scenes, semantic graphs, or interactions, as well as papers that mainly focused on image processing, AI, and deep learning. A total of 783 articles were excluded, leaving 87 articles at the end of the screening process. Subsequently, we excluded 19 articles that didn't provide enough information about their ontology or architecture, 7 state-of-the-art studies and thesis, 10 articles that only used semantics for visual annotation, and 6 articles that were off-topic (e.g. semantic representation of user emotions or navigation with voice commands). Finally, we ended up with 45 articles from the eligibility steps. We did not add any papers to our search, as we considered that we already had a sufficient number of papers for our analyses. Therefore, we retained a total of 45 articles. The full screening and eligibility process is outlined in Figure 1.

4 Result

To be able to answer the research question, we analyzed the 45 papers selected according to different criteria. In the following section, we present the results of our classification of these articles along different axes in terms of devices, semantic language, data represented, field and motivation. For each category, an overview will be presented, followed by a highlighting of research papers that are representative or have an impact on the category. The classification process was derived from the analyzed documents to obtain the most appropriate categories that can potentially be applied even to future work. Separating articles according to platform criteria has been effective for the selection process, as it suggests the inclusion of XR in these systems. The semantic language and data type represented categories answer the questions of what and how we represent VEs. Meanwhile, the field and motivation give more information about where, when and why we need to represent VEs. With these categories, we hope to provide an overview of the representation and use of semantic data in VEs. The number of articles in each category corresponds to the total number of articles because they have been counted uniquely.

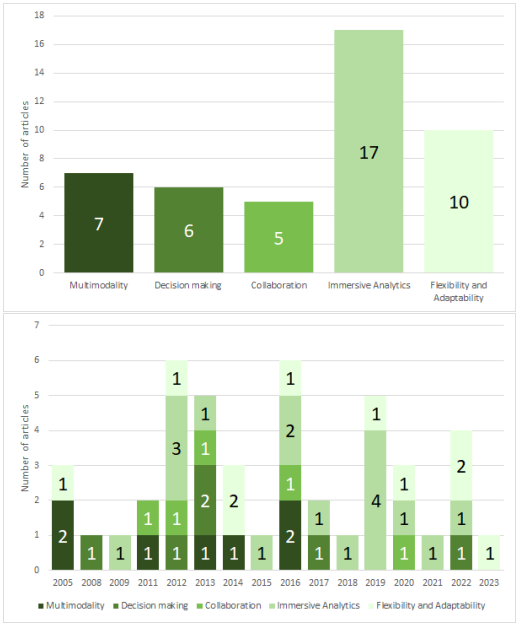


Figure 3: Number of articles classified by motivation (left), and grouped by date (right).

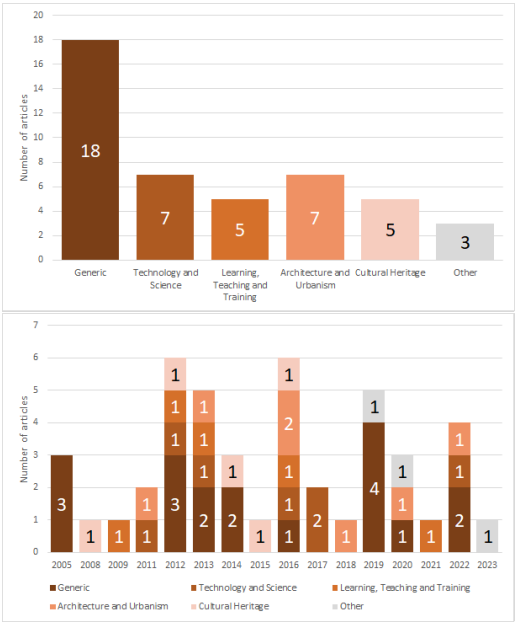


Figure 4: Number of articles classified by application field (left), and grouped by date (right).

4.1 Number of Publications

Analysis of scientific articles by year reveals notable variability between each year, particularly between 2012 and today. Over this period, the combined percentage of published works represents 84.4% of the total number of articles selected, indicating a significant concentration of research activity over the last decade (probably thanks to the stability of knowledge representation since the arrival of SPARQL 1.1 in 2013, and the emergence of VR/AR technology in those years). Regular spikes can be justified by the fact that semantics in virtual environments is a niche field, heavily dependent on the emergence of new technologies and representation techniques (in both semantic and immersive domains), but also by the fact that much semantic work in virtual environments is based on the use of artificial intelligence and image processing for recognition in particular, except that these papers were not selected because of the analysis criteria presented in Section 3.

4.2 Motivation

In this category, we are focusing on the rationale behind the use of semantics in virtual environments. To classify them correctly, we have grouped them in the following ways:

- **Multimodality** involves the use of various modes of input (interaction) and output (stimuli). [32, 45, 21, 16, 5, 8, 41]
- **Decision making** involves decision support systems (DSS) to aid users in making informed choices based on semantic information present in the virtual environment, allowing for more effective decision-making processes. [10, 9, 22, 23, 52, 7]
- **Collaboration** focuses on coordinating the experiences of multiple users to achieve a task. [20, 1, 14, 31, 53]
- **Immersive Analytics** involve presenting data for consultation or analysis. [2, 33, 44, 42, 30, 18, 51, 39, 29, 26, 47, 27, 34, 19, 11, 12, 17]
- **Flexibility and Adaptability** concerns data systems capable of adapting to different situations, such as different platforms or applications. [50, 24, 28, 48, 49, 43, 40, 13, 46, 35]

An important point to note is the absence of paper motivated by multimodality from 2016 (almost 10 years ago), despite the correct presence of 25% on all papers before this date. This decline seems to be due to solutions using specific algorithms [6] or artificial intelligence [25] instead of semantics. Therefore, our criteria excluded recent research on multimodal interaction from our selection. However, I'm convinced that such approaches will reach their limits in the future, due to the sheer quantity of contextual information to be processed (which will be from different domains for reasons of flexibility). Moreover, processing all the context locally raises the question of autonomy, hence the interest in finding an approach using knowledge graphs to do all the computation on a third-party server.

We can also see that the majority of papers are motivated by data consultation and analysis with 37.8% of the total count. It's probably because immersive technologies are extremely effective in exploring large datasets and because SPARQL/RDF/OWL technologies are currently very efficient for querying large datasets, due to their deduction capabilities and execution time.

4.3 Application Field

- **Generic** studies focus on fundamental principles and general methodologies. [10, 9, 50, 32, 44, 42, 1, 18, 24, 43, 26, 11, 13, 21, 16, 52, 17, 8]
- **Technology and Science** studies delve into specific scientific disciplines such as biology, chemistry, physics, engineering, and related fields. [45, 29, 23, 47, 41, 49, 33]
- **Learning, Teaching and Training** focuses on educational aspects, including instructional strategies, learning theories, and training methodologies. [2, 14, 51, 53, 12]
- **Architecture and Urbanism** explores the design and planning of urban areas, buildings, and landscapes. [5, 40, 19, 20, 28, 22, 46]

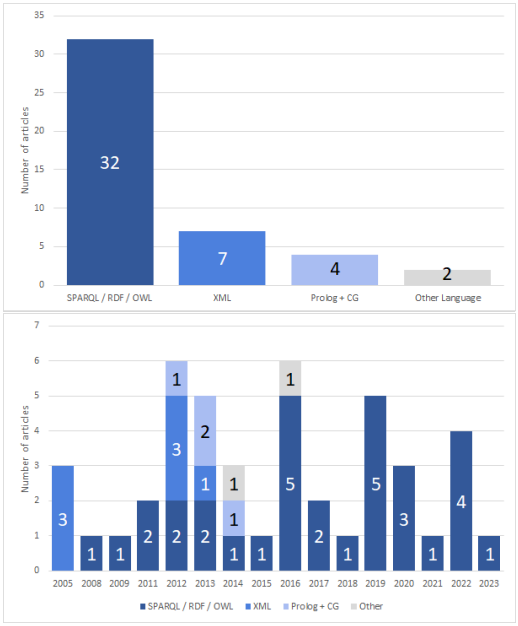


Figure 5: Number of articles classified by semantic technology (left), and grouped by date (right).

- **Cultural Heritage** investigates the preservation, interpretation, and promotion of cultural heritage sites, artefacts, and traditions. [31, 30, 48, 34, 7]

We can see that there is a large amount of research in the generic category, which constitutes 40% of the total selected articles from 7 categories. This huge amount is justified by the advantage of semantic flexibility achieved through the concept of "open world". This semantic flexibility allows for the extension of data and ontologies to infinity, enabling the creation of highly versatile and adaptable systems based on generic concepts that will become increasingly specific as the ontology evolves.

Another reason why there are so many articles in generic field is that semantics and VR/AR are two specific fields characterized by a high degree of skill complexity required. These fields are not often interconnected, and if on top of this, they have to be applied to an application field which may also require additional skills (e.g. expert or engineering), then there are very few people capable of doing anything other than a generic approach. We can also add that semantics and virtual reality are two fields that can require heavy engineering work, requiring more development than research and tending to be devalued, which may be one of the main reasons for the absence of articles in certain years. It's also worth noting that in recent years, semantic data management software has greatly improved, making it much easier to use than ever before. This was often an initial obstacle to semantic exploration.

Three special cases didn't fit in any application field. In 2019, Kanak et al. [27] explored the visual representation of food to help decision. In 2020, Petrovic N. [39] developed a system for generating artistic environments linked to music. In 2023, Maik et al. [35] focused on visualizing data related to warehouse storage.

The only thing we can deduce from this is that the semantic is mostly used generically, presumably to address problems that can be generalized across multiple fields simultaneously, or for other reasons that we will explore later in Figure 7.

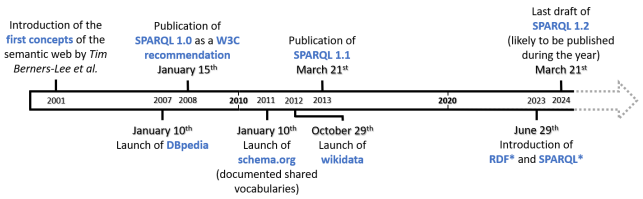


Figure 6: Introduction by Berners-Lee et al. [3] and evolution of the web semantic.

4.4 Semantic Technology

In this category, we classify articles by the semantic technology used to satisfy their motivation. Figure 5 shows an overwhelming presence of the semantic technology SPARQL/RDF/OWL, which has been a W3C recommendation since 2008. This presence is even more impressive from 2015 onward, attesting to a presence of almost 100%. This omnipresence is undoubtedly due to the arrival of SPARQL 1.1 in 2013, establishing it as the primary standard for knowledge representation.

Prolog is a 50-year-old French language for solving logic problems, whether for AI or knowledge bases. The disappearance of Prolog in 2014 is attributed to the obsolescence of Prolog itself. With the latest version dating back to January 2013, Prolog gradually lost relevance in subsequent years.

In contrast, XML's presence is discussed differently. XML serves as a serialization format, unlike Prolog and SPARQL/RDF/OWL acting as a data modelling language, making it impossible to manage rules and inferences. This distinction suggests that XML's role in the category may be more nuanced, influenced by factors beyond just semantic representation. But you can see that XML disappeared just as SPARQL 1.1 appeared, and that's actually because SPARQL/RDF/OWL supports XML for data management. There were also two special cases in 2014 and 2016. In 2014, Wiebusch and Latoschik [50] used Scala to model context. This is a very special case since Scala is not a language for representing data, but rather for scalable method programming. Nevertheless, this use is justified here by the fact that their motivation was to have flexibility between different types of rendering. In 2016, Torre and Celik [46] used The Global Public Inclusive Infrastructure (GPII) to virtualize real-world objects for interface adaptation. This approach is atypical given the use of a very specific service; nevertheless, it makes sense given that this service is designed for accessibility.

Overall, the observed trends highlight the dynamic nature of semantic technologies and their adaptation to ever-changing technological landscapes. While some technologies may fade into obsolescence due to technological advancements, others, like SPARQL/RDF/OWL, continue to thrive and shape the way we represent and interact with knowledge in digital environments.

4.5 Motivation Context

We cross-referenced the data between motivation and field of application.

In figure 7, we can see research on multimodality is only in the generic, expert, and architecture/urbanism (only 1 paper) application fields. The reason for this is that multimodality is much less explored than the other motivations, making it easier to apply to exploratory application domains such as generic ones or to extremely precise problems as might be found in expert domains.

We can also see that only collaboration and data analysis/consultation touch on the subjects of learning/teaching/training. The reason is that this field of application cannot be applied too broadly to other contexts. For flexibility and generic motivations,

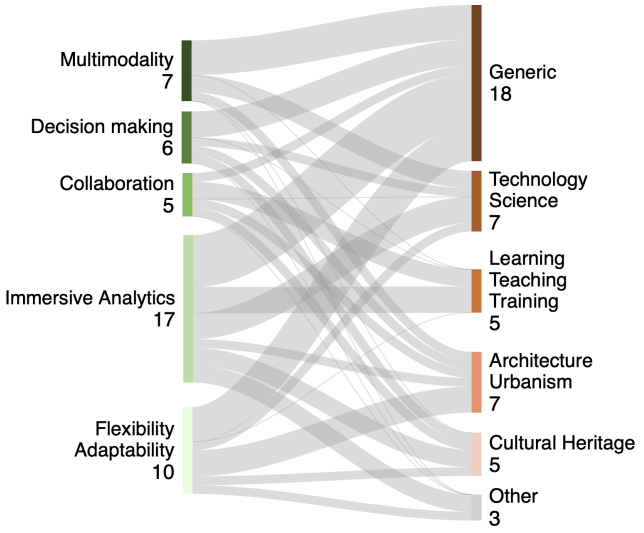


Figure 7: Sankey flow diagram from Motivation at left to Fields and Application at right

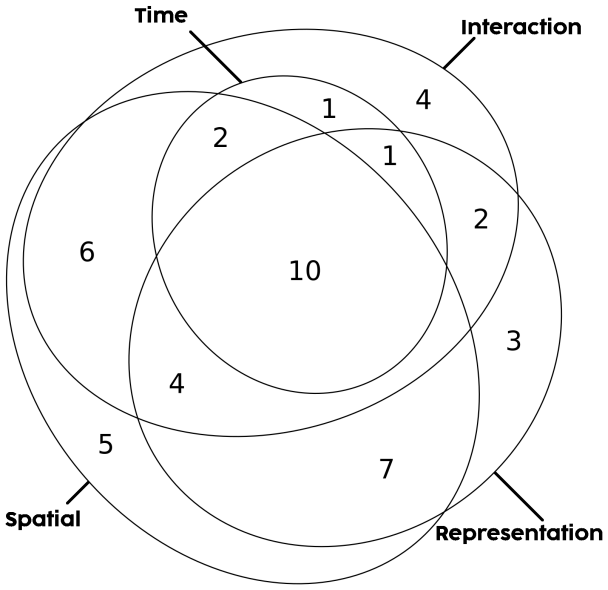


Figure 8: Number of articles that represented a specific type of data to achieve their motivation.

it is impossible because there are too many possible learning domains. This application field aims to learn, train, or develop skills that will be applicable in real-life conditions, but there are few real-life situations where we need to use multimodal interaction or be in the presence of systems that help with decision-making, so it is difficult to create a link between learning/teaching/training and these two motivations.

4.6 Data type

In this category, we are focusing on the rationale behind the use of semantics in virtual environments. To classify them correctly, we have grouped them in the following ways:

- **Spatial** involves representing the object's transformation in space, encompassing its position, rotation, and size. [10, 9, 31, 50, 44, 42, 14, 22, 30, 1, 18, 45, 24, 28, 48, 51, 43, 39, 26, 23, 47, 35, 34, 19, 11, 40, 21, 16, 52, 5, 17, 8, 41, 7]
- **Representation** refers to depicting an object, typically in its 3D form. [9, 53, 2, 33, 46, 42, 22, 1, 18, 45, 24, 28, 48, 51, 39, 29, 26, 23, 47, 27, 35, 34, 19, 11, 16, 17, 7]
- **Interaction** pertains to representing events that have occurred, often through interaction logging. [10, 9, 53, 2, 33, 50, 32, 18, 45, 24, 28, 48, 49, 43, 39, 20, 26, 23, 47, 35, 19, 40, 12, 13, 21, 16, 5, 8, 41, 7]
- **Time** involves the representation of the passage of time. [33, 50, 18, 45, 24, 28, 20, 26, 23, 47, 19, 16, 5, 7]

Many papers in the field commonly utilize a combination of spatial representation, object depiction, interaction logging, and time representation to achieve motivation for the use of semantics in virtual environments, with nearly 20% of studies incorporating these elements collectively.

Time is never represented alone; rather, it is often paired with at least two other forms of data. This is because time is a type of data that plays a crucial role in forming connections between other data types, especially to know the content at a specific instant. Therefore, it is recommended (and even necessary) to incorporate time into the representation of content with different types of data.

5 Discussion

In this section, building upon the various classifications outlined in the previous section, our goal is to address individually the research questions raised in the introduction.

5.1 RQ1: What are the advantages of representing the content of VEs?

In literature, we have discovered that there are many benefits to representing content in VEs. We expected to find work motivated by the multimodality of interaction in VEs, which is our goal. For example, Quasim et al. [41] modelled molecular chains in knowledge graphs, to generate commands for editing them (colouring or displaying/hiding parts of the molecule), which would have made it possible to generate multimodal commands in future work.

On the other hand, Trellet et al. [47] have also worked with semantics in the field of molecular biology, but this time with the aim of realizing visual analytics. They designed a system that enabled in-depth analysis of the 3D conformation of a molecule over time, with two distinct visualizations: a 2D visualization (point cloud) that allowed different types of selection (temporal entity, chain, residue, atom), and an explorable 3D visualization of what had been selected.

There are also advantages to collaboration, as shown by Zinnikus et al. [53] when they created their configurable collaborative platform for 3D scene editing, evaluation, and analysis. Additionally, there are benefits for decision support system, as Zheng et al. [52] in their AR project where they were able to provide suggestions to the user based on the current context (e.g., close the window of a room if the temperature exceeds a threshold; or analyze water consumption).

Flexibility is the last major point in representing the content of VEs. Eckstein et al. [13] use the advantages of knowledge graphs (high interoperability and rule-based inference) to create links between data from many different contexts (room temperature, user emotion, user tracking) to automatically control the ambient temperature of the room in which the user is located.

It can also be observed that in all the papers selected, we very

S=Spatial — R=Representation — I=Interaction — T=Time

Reference	Author	Date	Application Field: Generic	Technology and Science	Learning Teaching Training	Architecture Urbanism	Cultural Heritage	Other
Multimodality (7)								
[32]	Latoshick	2005	I					
[21]	Gutierrez et al.	2005	SI					
[45]	Takashide et al.	2011		SRIT				
[41]	Qasim et al.	2013		SI				
[8]	Dennemont et Bouyer	2015	SI					
[16]	Fischbash et al.	2016	SRIT					
[5]	Briochi et al.	2016				SIT		
Decision making (6)								
[7]	Chu et Li	2008					SRIT	
[9]	Dennemont et al.	2012	SRI					
[10]	Dennemont et al.	2013	SI					
[22]	Harkouken et al.	2013				SR		
[23]	Havard et al.	2017		SRIT				
[52]	Zheng et al.	2022	S					
Collaboration (5)								
[20]	Fulop	2011				IT		
[31]	Kovachev et al.	2012					S	
[53]	Zinnikus et al.	2013			RI			
[14]	Felemban et al.	2016			S			
[1]	Adwernat et al.	2020	SR					
Visual and Immersive Analytics (17)								
[51]	Xuesong et al.	2009			SR			
[42]	Shi et al.	2012	SR					
[33]	Lee et al.	2012		IRT				
[44]	Takalo-Mattila et al.	2013	S					
[34]	Lombardo et al.	2015					SR	
[47]	Trellet et al.	2016		SRIT				
[30]	Kim et al.	2016					S	
[29]	Katsaros et Keramopoulos	2017		R				
[19]	Flotynski et Sobocinski	2018				SRIT		
[18]	Flotynski et Nowak	2019	SRIT					
[11]	Djordjevic et al.	2019	SR					
[17]	Flotynski et al.	2019	SR					
[12]	Dris et al.	2019			I			
[27]	Kanak et al.	2019						R
[39]	Petrovic	2020						SRI
[2]	Antoniou et al.	2021			IR			
[26]	Jaimini et al.	2022	SRIT					
Flexibility and Adaptability (10)								
[24]	Heumer et al.	2005	SRIT					
[43]	Silva et al.	2012	SI					
[50]	Wiebusch et Latoschik	2014	SIT					
[48]	Walczak et al.	2014					SRI	
[46]	Torre et Celik	2016				R		
[13]	Eckstein et al.	2019	I					
[40]	Petrovic et Tosic	2020				SI		
[49]	Weigand et Fay	2022		I				
[28]	Kanak et al.	2022				SRIT		
[35]	Maik et al.	2023						SRI

Table 1: References classified by Authot, Date, Motivation, Application Field and Data Type

often find the same type of data to represent a scene (spatial, representation, interaction, time), whereas very different themes are addressed. This proves that there's a lot of generic work to be done to have a system that can support all these motivations, and moreover make data from several different application fields interoperable.

In summary, there is a lot to be said for taking advantage of data semantics in VEs. In particular, thanks to its ability to communicate information between different devices, whether between several users for collaboration or between different trackers for decision support systems and data flexibility. We can also take advantage of reasoning capabilities to perform visual analytics on very specific tasks, such as molecular data analysis, or to generate commands quickly for multimodal interaction purposes.

5.2 RQ2: What are the advantages of linking interactions with the content of a VE?

Representing the context of interaction in virtual environments (VEs) offers several key advantages. Firstly, it enables a more accurate, context-sensitive interpretation of user commands. By understanding not only the user's gestures or voice commands but also the context in which these interactions occur, systems can provide more appropriate and natural responses. For example, a system capable of recognizing that the user is pointing at a specific object while giving a voice command can better interpret this command in terms of the targeted object as shown by Latoschik in 2005 [32]

Secondly, the representation of interaction context improves the ability of systems to manage multimodal interactions in real-time. The late fusion of contextual data, where information from different modalities (gestures, speech, gaze) is integrated at a semantic level, creates more fluid and natural interactions. This is essential in virtual environments where users expect immersive interactions that mimic real-world interactions. Furthermore, semantics make it possible to combine different motivations at the same time that cover similar issues, as Martin et al. [36] did by combining multimodal interaction and collaboration.

What's more, representing context enables greater personalization of the user experience. By capturing contextual information such as user preferences, interaction history and the current state of the VE, systems can dynamically adapt the interfaces and content offered, improving user engagement and satisfaction. For example, Maik et al. [35] have presented a system that enables interface adaptation in Virtual Reality based on the user's context about the virtual scene, using rules. For example, if the user has eyesight problems, the interface can be adapted by making it larger or the colours brighter, or more generally, if the user tends not to look around at 360°, the interface can be restricted to be placed more often in front of the user. There's also an interest in object gripping as if the user is quite stationary or has difficulty reaching buttons or interface elements, these will naturally come closer to him or her.

5.3 RQ3: What contributions can the use of knowledge graphs and rule-based reasoning make to interaction management in VEs?

The use of knowledge graphs and rule-based reasoning significantly improves interaction management in virtual environments (VEs) by providing structured and contextually enriched frameworks for decision-making and interpretation of user interactions.

Knowledge graphs enable VEs to maintain rich contextual information about objects, users and interactions within the environment. This enables the system to understand user actions based on comprehensive semantic relationships. For example, Flotynski and Nowak [18] have made a car exhibition in VR where they could answer questions such as *"How much time do customers spend to watch the particular cars outside and inside?"* or *"What are the most popular colours selected by customers for the particular cars?"*

By integrating various types of data (visual, auditory, textual) into a unified graph structure, knowledge graphs facilitate seamless multimodal interactions. In other words, the commands used, whether vocal or gestural, can be interpreted more precisely in the context of the semantic structure of the environment, enabling new commands to be generated (multimodal fusion [37]). For example, Qasim et al. [41] explore an approach in the application field of Molecular Biology, where they could select and change the properties (colour and visibility) of a molecule's part by using uniquely vocal commands, but they point out that in future work, this approach could be used to manage multimodal interaction commands.

Knowledge graphs can store detailed user profiles and preferences, enabling VEs to personalize interactions and dynamically adapt to individual users. For example, if a user frequently uses certain tools or objects, the system can prioritize them in future interactions, improving user experience and efficiency. For example, as mentioned in the previous question, Maik et al. [35] have created a system for obtaining and inferring user preferences to adapt Virtual Reality interfaces.

Rule-based reasoning provides a clear and explicit framework for managing interactions. Defining specific rules for interpreting user actions and system responses, guarantees consistent, predictable behaviour. For example, Latoschik [32] has designed a system for assigning verbs ("turn", "rotate") to actions (rotating an object) with the object information (What type? Is it rotatable?) in focus.

Rule-based systems are highly adaptable, making it possible to add new rules or modify existing ones to meet new functionalities or user needs. This flexibility supports the evolution of VEs as user requirements and technologies change. This is one of the reasons why there is so much research in the generic application field in semantics in VEs, as shown in Figure 4.

By providing explicit rules for interaction, this approach reduces ambiguity and improves the accuracy of interpretations of user actions/states, whether for a human or a machine. It is much simpler to understand the knowledge that "user X is looking at object Y at time T" than to have information on the direction in which user X is looking, and the position of object Y, each at time T, which is obviously difficult for a human to interpret and analyze. For example, from information such as the user's height, concentration score or anxiety score (each of which is a value), Maik et al. [35] have been able to determine, using rules, much more explicit knowledge, classifying users into different categories (e.g. ShortUser, DyslexicUser and AnxiousUser) to be able to adapt the interface correctly to the user.

Finally, the use of knowledge graphs has a performance advantage. The graphs take care of all the reasoning and calculations, which are hosted on servers external to the execution environment, leaving more resources for the client's device. This is an excellent thing in the field of Virtual and Augmented Reality, where there is already a trade-off between performance, autonomy and device weight, resulting in devices with performance shortcomings in certain use cases. It's also worth noting that some headsets feature Augmented Reality functionality, enabling real objects to be tracked and stored using semantic representation like virtual objects presented throughout this article.

6 Conclusion

In this article, we conducted a literature review on the use of semantics in virtual environments (VEs). Our analysis focused on 45 selected articles. We classified these articles according to different axes: motivation, application domain, semantic language and data types. We went deeper into each article to reveal different examples of use, and technical solutions for representing the virtual environment, to deduce guidelines for future research in the field of semantics in VEs.

The study reveals that the integration of semantic technologies in virtual spaces considerably improves interpretation and interaction capabilities within the VEs. By using knowledge graphs and rule-based reasoning, virtual spaces can have a relational representation of their scene, enabling better personalization of the user experience (such as adapting the interface or the actions it will undertake), as well as better interaction management thanks to the notion of interaction context observer, making semantic links between content and interaction events. We also found that most motivations had already explored in depth the advances offered by SPARQL 1.1, deployed in 2013, except multimodality. For the latter, solutions more oriented towards functionality or artificial intelligence (early fusion) were favoured. Nevertheless, it would be interesting to explore the use of knowledge graphs to manage multimodal interactions, not least because the latest research on the use of semantics to accomplish this task dates back to 2016. Some articles consider this motivation as an avenue for future work, demonstrating a real interest in being explored. It's worth noting, however, that despite a great deal of interest and multiple avenues of development, both in terms of motivation and field of application, semantics and Virtual and Augmented Reality are two technically complex fields, which can make their progress together very slow due to the small community in this intersection of the fields of knowledge representation and virtual reality. Therefore, an increasing amount of future research is to be expected, especially when we know that SPARQL 1.1 (updated in 2013) will receive an update this year after 11 years to become SPARQL 1.2.

REFERENCES

[1] S. Adwernat, M. Wolf, and D. Gerhard. An ontology-based concept to support information exchange for virtual reality design reviews. In *Product Lifecycle Management Enabling Smart X: 17th IFIP WG 5.1 International Conference, PLM 2020, Rapperswil, Switzerland, July 5–8, 2020, Revised Selected Papers 17*, pp. 270–284. Springer, 2020. 2, 4, 6, 7

[2] P. E. Antoniou, E. Chondrokostas, C. Bratsas, P.-M. Filippidis, and P. D. Bamidis. A medical ontology informed user experience taxonomy to support co-creative workflows for authoring mixed reality medical education spaces. In *2021 7th International Conference of the Immersive Learning Research Network (iLRN)*, pp. 1–9. IEEE, 2021. 1, 4, 6, 7

[3] T. Berners-Lee, J. Hendler, and O. Lassila. A new form of web content that is meaningful to computers will unleash a revolution of new possibilities. *Scientific american*, 284(5):34–43, 2001. 5

[4] R. A. Bolt. "put-that-there" voice and gesture at the graphics interface. In *Proceedings of the 7th annual conference on Computer graphics and interactive techniques*, pp. 262–270, 1980. 1

[5] G. Briochi, M. Colombetti, M. D. Hina, A. Soukane, and A. Ramdane-Cherif. Techniques for cognition of driving context for safe driving application. In *2016 IEEE 15th International Conference on Cognitive Informatics & Cognitive Computing (ICCI* CC)*, pp. 388–397. IEEE, 2016. 2, 4, 6, 7

[6] L. Cao, H. Zhang, C. Peng, and J. T. Hansberger. Real-time multimodal interaction in virtual reality-a case study with a large virtual interface. *Multimedia Tools and Applications*, 82(16):25427–25448, 2023. 4

[7] Y.-L. Chu and T.-Y. Li. Using pluggable procedures and ontology to realize semantic virtual environments 2.0. In *Proceedings of The 7th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and Its Applications in Industry*, pp. 1–6, 2008. 4, 5, 6, 7

[8] Y. Dennemont and G. Bouyer. Un moteur de raisonnement sémantique pour des interactions 3d sensibles au contexte. *Rev. d'Intelligence Artif.*, 29(6):705–736, 2015. 4, 6, 7

[9] Y. Dennemont, G. Bouyer, S. Otmane, and M. Mallem. 3d interaction assistance through context-awareness. In *2012 IEEE Virtual Reality Workshops (VRW)*, pp. 103–104. IEEE, 2012. 4, 6, 7

[10] Y. Dennemont, G. Bouyer, S. Otmane, and M. Mallem. 3d interaction assistance in virtual reality: a semantic reasoning engine for context-awareness. In *Context-Aware Systems and Applications: First International Conference, ICCASA 2012, Ho Chi Minh City, Vietnam, November 26-27, 2012, Revised Selected Papers 1*, pp. 30–40. Springer, 2013. 4, 6, 7

[11] L. Djordjevic, N. Petrovic, and M. Tosic. Ontology based approach to development of augmented reality applications. In *2019 27th Telecommunications Forum (TELFOR)*, pp. 1–4. IEEE, 2019. 4, 6, 7

[12] A.-S. Dris, F. Lehericey, V. Gouranton, and B. Arnaldi. Openbim based ive ontology: an ontological approach to improve interoperability for virtual reality applications. In *Advances in Informatics and Computing in Civil and Construction Engineering: Proceedings of the 35th CIB W78 2018 Conference: IT in Design, Construction, and Management*, pp. 129–136. Springer, 2019. 4, 6, 7

[13] B. Eckstein, F. Niebling, and B. Lugrin. Reflected reality: A mixed reality knowledge representation for context-aware systems. In *2019 11th International Conference on Virtual Worlds and Games for Serious Applications (VS-Games)*, pp. 1–4. IEEE, 2019. 4, 6, 7

[14] S. Felemban, M. Gardner, and V. Callaghan. An event detection approach for identifying learning evidence in collaborative virtual environments. In *2016 8th Computer Science and Electronic Engineering (CEEC)*, pp. 42–47. IEEE, 2016. 4, 6, 7

[15] N. Férey, J. Nelson, C. Martin, L. Picinali, G. Bouyer, A. Tek, P. Bourdot, J.-M. Burkhardt, B. F. Katz, M. Ammi, et al. Multisensory vr interaction for protein-docking in the corsaire project. *Virtual Reality*, 13:273–293, 2009. 2

[16] M. Fischbach, D. Wiebusch, and M. E. Latoschik. Semantics-based software techniques for maintainable multimodal input processing in real-time interactive systems. In *2016 IEEE 9th Workshop on Software Engineering and Architectures for Realtime Interactive Systems*

(SEARIS), pp. 1–6. IEEE, 2016. 4, 6, 7

[17] J. Flotyński, D. Brutzman, F. G. Hamza-Lup, A. Malamos, N. Polys, L. F. Sikos, and K. Walczak. The semantic web3d: towards comprehensive representation of 3d content on the semantic web. In *2019 International Conference on 3D Immersion (IC3D)*, pp. 1–9. IEEE, 2019. 4, 6, 7

[18] J. Flotyński and A. Nowak. Annotation-based development of explorable immersive vr/ar environments. In *2019 International Conference on 3D Immersion (IC3D)*, pp. 1–9. IEEE, 2019. 4, 6, 7, 8

[19] J. Flotyński and P. Sobociński. Logging interactions in explorable immersive vr/ar applications. In *2018 International Conference on 3D Immersion (IC3D)*, pp. 1–8. IEEE, 2018. 4, 6, 7

[20] I. M. Fülöp. Extending intelligent collaboration using semantic information in the virca system. In *2011 IEEE 9th International Symposium on Applied Machine Intelligence and Informatics (SAMI)*, pp. 237–242. IEEE, 2011. 4, 6, 7

[21] M. Gutiérrez, D. Thalmann, and F. Vexo. Semantic virtual environments with adaptive multimodal interfaces. In *11th International Multimedia Modelling Conference*, pp. 277–283. IEEE, 2005. 2, 4, 6, 7

[22] K. Harkouken, N. Sabouret, and J.-Y. Donnart. An ontology for semantic representation of an urban virtual environment. In *The International Conference on Artificial Intelligence*, 2013. 4, 6, 7

[23] V. Havard, B. Jeanne, X. Savatier, and D. Baudry. Inoovas-industrial ontology for operation in virtual and augmented scene: The architecture. In *2017 4th International Conference on Control, Decision and Information Technologies (CoDIT)*, pp. 0300–0305. IEEE, 2017. 1, 4, 6, 7

[24] G. Heumer, M. Schilling, and M. E. Latoschik. Automatic data exchange and synchronization for knowledge-based intelligent virtual environments. In *IEEE Proceedings. VR 2005. Virtual Reality, 2005.*, pp. 43–50. IEEE, 2005. 4, 6, 7

[25] J. Izquierdo-Domenech, J. Linares-Pellicer, and J. Orta-Lopez. Towards achieving a high degree of situational awareness and multimodal interaction with ar and semantic ai in industrial applications. *Multimedia Tools and Applications*, 82(10):15875–15901, 2023. 4

[26] U. Jaimini, T. Zhang, G. O. Brikis, and A. Sheth. imetaversekg: industrial metaverse knowledge graph to promote interoperability in design and engineering applications. *IEEE Internet Computing*, 26(6):59–67, 2022. 1, 4, 6, 7

[27] A. Kanak, İ. Arif, A. M. Karadeniz, and S. Ergün. Kebap: A semantic key enabling beverage and appetite platform enriched with virtual reality. In *2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)*, pp. 205–210. IEEE, 2019. 4, 5, 6, 7

[28] A. Kanak, İ. Arif, Ç. Terzibaş, Ö. F. Demir, and S. Ergün. Bimyverse: Towards a semantic interpretation of buildings in the city and cities in the universe. In *2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 1072–1077. IEEE, 2022. 1, 4, 6, 7

[29] A. Katsaros and E. Keramopoulos. Farmar, a farmer's augmented reality application based on semantic web. In *2017 South Eastern European Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)*, pp. 1–6. IEEE, 2017. 4, 6, 7

[30] H. Kim, T. Matuszka, J.-I. Kim, J. Kim, and W. Woo. An ontology-based augmented reality application exploring contextual data of cultural heritage sites. In *2016 12th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS)*, pp. 468–475. IEEE, 2016. 4, 5, 6, 7

[31] D. Kovachev, G. Aksakali, and R. Klamma. A real-time collaboration-enabled mobile augmented reality system with semantic multimedia. In *8th International Conference on Collaborative Computing: Networking, Applications and Worksharing (CollaborateCom)*, pp. 345–354. IEEE, 2012. 4, 5, 6, 7

[32] M. E. Latoschik. A user interface framework for multimodal vr interactions. In *Proceedings of the 7th international conference on Multimodal interfaces*, pp. 76–83, 2005. 1, 2, 4, 6, 7, 8

[33] K.-S. Lee, A. NurzidRosli, and G.-S. Jo. A method for automatically creating an interactive semantic video based on ar system. In *2012 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 2626–2631. IEEE, 2012. 4, 6, 7

[34] V. Lombardo, R. Damiano, and A. Lieto. Knowledge-based framework for the virtual communication of public art. In *2015 9th International Conference on Innovative Mobile and Internet Services in Ubiquitous Computing*, pp. 117–124. IEEE, 2015. 4, 5, 6, 7

[35] M. Maik, J. Flotyński, and K. Walczak. Knowledge-based adaptation of xr user interfaces for stock market data visualization. In *2023 International Conference on Cyberworlds (CW)*, pp. 94–101. IEEE, 2023. 4, 5, 6, 7, 8

[36] P. Martin, A. Tseu, N. Férey, D. Touraine, and P. Bourdot. A hardware and software architecture to deal with multimodal and collaborative interactions in multiuser virtual reality environments. In *The Engineering Reality of Virtual Reality 2014*, vol. 9012, pp. 68–83. SPIE, 2014. 2, 8

[37] S. Oviatt, P. Cohen, L. Wu, L. Duncan, B. Suhm, J. Bers, T. Holzman, T. Winograd, J. Landay, J. Larson, et al. Designing the user interface for multimodal speech and pen-based gesture applications: State-of-the-art systems and future research directions. *Human-computer interaction*, 15(4):263–322, 2000. 2, 8

[38] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, et al. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *International journal of surgery*, 88:105906, 2021. 2

[39] N. Petrovic. Digital consumption as art expression: Ontological approach to vaporwave generation. In *2020 Zooming Innovation in Consumer Technologies Conference (ZINC)*, pp. 118–122. IEEE, 2020. 4, 5, 6, 7

[40] N. Petrovic and M. Tosic. Ontology-driven generation of interactive 3d worlds. In *2020 19th International Symposium INFOTEH-JAHORINA (INFOTEH)*, pp. 1–5. IEEE, 2020. 4, 6, 7

[41] M. Qasim, G. Bouyer, and N. Férey. Using conceptual graphs to embed expert knowledge in vr interactions: Application to structural biology. In *5th Joint Virtual Reality Conference–JVRC 2013*, 2013. 1, 2, 4, 6, 7, 8

[42] M. Shi, H. Cai, and L. Jiang. An approach to semi-automatic semantic annotation on web3d scenes based on an ontology framework. In *2012 12th International Conference on Intelligent Systems Design and Applications (ISDA)*, pp. 574–579. IEEE, 2012. 4, 6, 7

[43] E. Silva, N. Silva, H. Paredes, P. Martins, B. Fonseca, and L. Morgado. Development of platform-independent multi-user choreographies for virtual worlds based on ontology combination and mapping. In *2012 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, pp. 149–152. IEEE, 2012. 4, 6, 7

[44] J. Takalo-Mattila, J. Kiljander, and J.-P. Soininen. Advertising semantically described physical items with bluetooth low energy beacons. In *2013 2nd Mediterranean Conference on Embedded Computing (MECO)*, pp. 211–214. IEEE, 2013. 4, 6, 7

[45] R. Takashide, Y. Chigusa, and S. Takahira. Approaching the measurement about human behavior and environmental behavior by" guideware". application possibility of "guideware" (report 1). In *SICE Annual Conference 2011*, pp. 2084–2089. IEEE, 2011. 4, 6, 7

[46] I. Torre and I. Celik. A model for adaptive accessibility of everyday objects in smart cities. In *2016 IEEE 27th annual international symposium on personal, indoor, and mobile radio communications (PIMRC)*, pp. 1–6. IEEE, 2016. 4, 5, 6, 7

[47] M. Trellet, N. Ferey, M. Baaden, and P. Bourdot. Interactive visual analytics of molecular data in immersive environments via a semantic definition of the content and the context. In *2016 Workshop on Immersive Analytics (IA)*, pp. 48–53. IEEE, 2016. 1, 4, 6, 7

[48] K. Walczak, D. Rumiński, and J. Flotyński. Building contextual augmented reality environments with semantics. In *2014 International Conference on Virtual Systems & Multimedia (VSMM)*, pp. 353–361. IEEE, 2014. 4, 5, 6, 7

[49] M. Weigand and A. Fay. Creating virtual knowledge graphs from software-internal data. In *IECON 2022–48th Annual Conference of the IEEE Industrial Electronics Society*, pp. 1–6. IEEE, 2022. 4, 6, 7

[50] D. Wiebusch and M. E. Latoschik. A uniform semantic-based access model for realtime interactive systems. In *2014 IEEE 7th Workshop on Software Engineering and Architectures for Realtime Interactive Systems (SEARIS)*, pp. 51–58. IEEE, 2014. 4, 5, 6, 7

[51] W. Xuesong, Z. Mingquan, and F. Yachun. Building vr learning en-

vironment: an ontology based approach. In *2009 First International Workshop on Education Technology and Computer Science*, vol. 3, pp. 160–165. IEEE, 2009. 1, 4, 6, 7

[52] M. Zheng, X. Pan, N. V. Bermeo, R. J. Thomas, D. Coyle, G. M. O'Hare, and A. G. Campbell. Stare: Semantic augmented reality decision support in smart environments. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 630–631. IEEE, 2022. 4, 6, 7

[53] I. Zinnikus, X. Cao, M. Klusch, C. Krauss, A. Nonnengart, T. Spieldenner, and P. Slusallek. A collaborative virtual workspace for factory configuration and evaluation. In *9th IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing*, pp. 353–362. IEEE, 2013. 4, 6, 7